Project NEXUS V1.5 is Now Open Source

Building the Future of Community Exchange — Together

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Marketplace | Donations | Identity Verification

Federation · Nexus + Komunitin + Credit Commons + TimeOverflow

11 Languages · RTL | Build-Time SEO | Stripe Payments

Passkeys · CSP · CORS

Dear Community Builders, Developers, and Changemakers,

Today marks a pivotal moment in our mission to reimagine how communities connect, exchange, and thrive.

> **We're releasing Project NEXUS V1.5 to the world — fully open source, built for everyone.** This updated announcement covers the major work shipped through Spring 2026: a standalone Marketplace, Donations, Identity Verification, a four-protocol Federation layer (Nexus, Komunitin, Credit Commons, TimeOverflow), 11 languages with right-to-left support, a build-time SEO pipeline, and a hardened payments and security layer.

What is Project NEXUS?

Project NEXUS is an enterprise-grade, multi-tenant community platform that puts people before profit. At its core is timebanking — a radical idea that every hour of your time is worth exactly one hour of someone else's, regardless of what you do.

But NEXUS goes far beyond timebanking. It is a complete ecosystem for building connected, engaged, and empowered communities — **and we're building the infrastructure to connect those communities across borders.**

What's Inside V1.5?

At a glance: **a timebanking engine, real-time messaging, AI-powered matching, multi-tenant architecture, a four-protocol federation layer (Nexus, Komunitin, Credit Commons, TimeOverflow), a Stripe payments layer (subscriptions, donations, marketplace payouts, identity verification), gamification, and a complete social platform in 11 languages** — all production-ready and fully Dockerized.

Core Platform

⏰ **Timebanking Engine** — full credit exchange system with wallet, transactions, and broker controls.

🏢 **Multi-Tenancy** — host unlimited communities on a single platform, each with its own branding and configuration.

🎯 **Smart Matching** — AI-powered matching with semantic embeddings, collaborative filtering, availability scheduling, and learned preferences.

💬 **Real-Time Messaging** — private conversations with Pusher WebSocket integration and live presence.

📱 **Progressive Web App + Native Mobile App** — install as a PWA on any device, or deploy natively via Capacitor (iOS & Android).

🌐 **Federation** — connect multiple communities into a network for cross-community exchange, shared listings, events, and messaging.

Member Experience

📋 **Service Listings** — post offers and requests, browse and smart-match listings.

🛒 **Marketplace** — a standalone classifieds module with Stripe Connect payouts, orders, seller profiles, and AI-powered reply suggestions.

💳 **Donations** — one-off and recurring donations via Stripe, with full dashboards and receipts for organisations.

🪪 **Identity Verification** — optional Stripe Identity flow (document + selfie + name/DOB matching) with a verified-member badge.

🔄 **Exchange Workflow** — structured service exchange lifecycle with broker approval.

👥 **Group Exchanges** — bulk community service exchanges.

📰 **Social Feed** — posts, comments, likes, polls, hashtags, voice messages, media attachments, link previews, and @mentions.

📸 **Stories** — ephemeral 24-hour photo and video stories with reactions, polls, and highlights.

🎉 **Presence, Events & Groups** — real-time online/offline status, community gatherings, and interest-based groups.

🤝 **Connections & Members Directory** — follow, connect, browse, filter, and discover.

🏆 **Gamification** — verification badges, journeys, XP, leaderboards, achievements, challenges, streaks, XP shop, community dashboard, and seasonal competitions.

💡 **Goals, Impact & Ideation** — personal goals with mentoring, community impact tracking, and innovation challenges with campaigns and voting.

🙋 **Volunteering** — opportunities, shifts, hours, check-ins, expenses, certificates, wellbeing monitoring, and emergency alerts — **now with a volunteer organisation time-credit wallet (member deposits, admin adjustments, and time-credit payouts to volunteers on approved hours).**

💼 **Jobs, Organisations & Sub-Accounts** — full recruitment with alerts and analytics, company profiles with org wallets, and household/family accounts.

⭐ **Reviews, Endorsements, Polls, Skills & Availability** — the trust, discovery, and scheduling primitives a community needs.

Content & Communication

📝 **Blog, Resources, Help Center & Custom Pages** — a full content layer for each tenant.

📰 **Newsletter System** — email campaigns with A/B testing, smart segments, geo targeting, recurring sends, templates, send-time optimisation, and full open/click analytics.

🤖 **AI Chat** — OpenAI-powered assistant for platform guidance.

📄 **Legal Hub** — versioned legal documents with acceptance gates and an audit trail.

🏥 **Impact Reports & Social Prescribing** — SROI analysis, member outcome reports, and tooling for community health integration workflows.

Trust & Reputation

✅ **Verification Badges & NexusScore** — member verification indicators and proprietary reputation scoring.

📊 **Personal Insights Dashboard** — individual engagement metrics, hours given/received, skills breakdown, and personalised recommendations.

🛡️ **Safeguarding & CRM** — flagged-content review workflow, incident reporting, safeguarding assignment tracking, and admin contact management with notes, tasks, tags, and timelines.

AI & Recommendation Engine

🔍 **Semantic Search** — Meilisearch-powered full-text search with synonyms, custom ranking, and per-tenant isolation.

🧠 **Collaborative Filtering & Semantic Embeddings** — OpenAI-powered content matching for listings, members, and requests.

📈 **EdgeRank Feed, MatchRank & CommunityRank** — Bayesian quality scoring with Wilson confidence intervals.

🔁 **Group Recommendations & Match Learning** — trending, affinity-based discovery, and a feedback loop that improves recommendations over time.

📊 **Algorithm Health Dashboard** — live admin monitoring and tuning of every ranking system.

Modern Tech Stack

- Frontend: React 18 + TypeScript + HeroUI + Tailwind CSS 4.
- Backend: Laravel 12 + PHP 8.2+.
- Database: MariaDB 10.11.
- Search: Meilisearch v1.7.
- AI: OpenAI `text-embedding-3-small`.
- Real-Time: Pusher WebSockets, Firebase Cloud Messaging.
- Infrastructure: Docker-ready, Redis caching, full PWA + Capacitor native app support.
- API: OpenAPI 3.0 specification with Swagger UI docs.

Built for Production

🔐 **Enterprise security** — CSRF, rate limiting, TOTP 2FA, WebAuthn passkeys, CSP nonces, a CORS allowlist, Form Request validation, email verification gates, and invite-code registration.

💳 **Stripe payments layer** — subscriptions, donations, marketplace (Connect), and identity verification, with idempotent webhook handling and deep money-flow test coverage.

🛡️ **GDPR compliance suite** — data requests, consent management, cookie consent, breach tracking, and a full audit log.

🚨 **Fraud & abuse detection** — automated suspicious activity alerts and content moderation.

🏢 **Enterprise RBAC** — role-based access control across 13+ modules with a full permission matrix.

♿ **WCAG 2.1 AA accessibility compliance.**

🌍 **11 languages** — English, Irish, German, French, Italian, Portuguese, Spanish, Dutch, Polish, Japanese, and **Arabic with full right-to-left layout.**

🔎 **SEO at build time** — 19+ public pages pre-rendered with Playwright, structured data, sitemaps, and Prerender.io fallback for dynamic routes.

📊 **Admin panel** — tenant configuration, algorithm controls, diagnostics, cron job monitoring, and email deliverability tracking.

⏳ **Tenant hierarchy** — parent-child tenant relationships with feature toggling per tenant.

🧪 **500+ PHPUnit tests** (money flow, webhooks, federation, groups, marketplace) plus Vitest suites — run on every commit.

📦 **Fully Dockerized.**

Global Federation: Connecting Timebanks Worldwide

V1.5 ships with native adapters for four federation protocols — Nexus (our own), the complete Komunitin specification (all 15 endpoints), the Credit Commons Protocol (11 endpoints, CEN-compatible mutual credit with a P→V→C→E→X transaction lifecycle), and **TimeOverflow**. Two-way

sync across 9 entity types (members, listings, events, groups, connections, volunteering, reviews, transfers, messages), event-driven push, reputation portability, and end-to-end integration test coverage.

The vision is simple but ambitious: **a global time bank where external federation partners can discover each other, exchange services, and trade time credits across platforms and across borders.**

Imagine a member in Dublin helping someone in São Paulo with web design, while a member in Tokyo teaches Irish students origami — all through interconnected timebanking platforms, all valuing every hour equally.

The Federation API enables:

🌍 **Cross-platform discovery** — find members and services on partner timebanking platforms globally.

🔄 **Interoperable time credit exchange** — trade time credits between different timebanking systems seamlessly.

🏘️ **Federation Neighborhoods** — geographically grouped clusters of federated communities for regional coordination.

📜 **Credit Agreements** — negotiated exchange rate terms between federated communities.

🤝 **External federation partnerships** — any timebanking platform worldwide can connect to the network via standardized API endpoints.

🛒 **Global marketplace** — a unified view of offers and requests spanning multiple platforms and communities.

🏅 **Trust & verification** — federated reputation and review systems that travel with members across platforms.

This isn't just about connecting NEXUS communities to each other — it is about building the **open infrastructure for a worldwide timebanking economy.**

We're inviting timebanking platforms, community currency projects, and mutual aid networks everywhere to join us in defining this standard together.

Why Open Source?

Because transformative technology should be accessible to everyone. Communities around the world deserve tools that respect their autonomy, protect their data, and adapt to their needs. By going open source under AGPL-3.0, Project NEXUS remains:

- ✅ **Free forever** — no vendor lock-in, no surprise fees.
- ✅ **Transparent** — every line of code is visible and auditable.
- ✅ **Customisable** — fork it, modify it, make it yours.
- ✅ **Community-driven** — shape the roadmap, contribute features, fix bugs together.

And by building the Federation API as an open standard, we're ensuring that **no single platform controls the global timebanking network.** The protocol belongs to everyone.

Get Started Today

GitHub Repository: github.com/jasperfordesq-ai/nexus-v1

Clone it, deploy it, break it, improve it. The full platform is yours to explore, including complete source code, a Docker Compose setup for instant local development, comprehensive documentation, migration scripts, the admin panel, production deployment guides, and the OpenAPI 3.0 specification.

Looking Ahead: Version 2

While V1 represents years of refinement and real-world use, we're already building the next generation.

Project NEXUS V2 is in early-stage development, exploring:

- Modern microservices architecture (ASP.NET Core 8 + React 18).
- Enhanced scalability for global networks.
- Advanced AI integration for matching and insights.
- **Extended Federation Protocol** — building on V1's Federation API with deeper cross-platform credit exchange and expanded global network capabilities.
- Next-generation UX patterns.

V2 is also fully open source from day one. Follow along at github.com/jasperfordesq-ai/api.project-nexus.net.

Join the Movement

Whether you're a developer, community organiser, timebanking platform operator, designer, documentation writer, or translator — **your voice matters, your contributions matter.**

- V1 Issues & Features: GitHub Issues
- V1 Discussions: GitHub Discussions
- V2 Development: V2 Repository

Early Adoption

Timebanks from around the world are now starting to pilot Project NEXUS as their community platform. From Ireland to the UK and beyond, real communities are using NEXUS to connect members, exchange services, and build social capital — and their feedback is shaping every release.

Acknowledgements & Contributors

Creator: Jasper Ford — primary author, copyright © 2024–2026.

Founders of hOUR Timebank CLG (the originating Irish timebank): Jasper Ford and Mary Casey.

Contributors: Steven J. Kelly (community insight & product thinking); Sarah Bird, CEO Timebanking UK (strategic guidance).

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Open Protocol Attribution

The federation layer implements three independent open protocols developed by the wider mutual-credit and timebanking community. Each protocol's specifications and reference implementations remain the work of their original authors; NEXUS provides adapter code only.

- **Komunitin** — JSON:API-based federation specification, independently authored by the Komunitin community.
- **Credit Commons Protocol** — recursive mutual-credit accounting standard used by the Credit Commons / CEN network.
- **TimeOverflow** — the open-source Spanish timebanking platform whose public API we interoperate with.

Sincere thanks to the **Komunitin**, **Credit Commons / CEN**, and **TimeOverflow** communities for the open specifications that make cross-platform federation possible.

Full attribution in NOTICE and CONTRIBUTORS.md.

The Future is Community-Owned — and Globally Connected

For too long, community platforms have been controlled by corporations optimising for profit over people. And for too long, timebanking communities have operated in isolation, unable to connect with kindred communities across the world.

Project NEXUS is different. It is built by the community, for the community, owned by everyone. With the Federation API, we're building the bridges that will connect timebanks into a **truly global network** — where every hour is equal, no matter where in the world it's given.

Today, we're not just releasing software. We're releasing a blueprint for how communities can thrive in the digital age — **together, across every border.**

> *"Every hour is equal. Every contribution matters.*
> *Every community deserves better. Everywhere."*

Welcome to Project NEXUS. Welcome home.